



MODULE 3 | PART 2

GEOMETRY

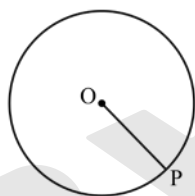
IOQM & RMO 2026

GEOMETRY II

Circle

The collection of all the points in a plane, which are at a fixed distance from a fixed point in the plane, is called a circle.

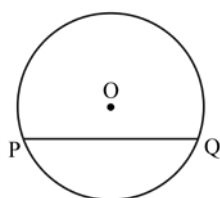
The fixed point is called the centre of the circle and the fixed distance is called the radius of the circle.



In figure, O is the centre and the length OP is the radius of the circle. So the line segment joining the centre and any point on the circle is called a radius of the circle.

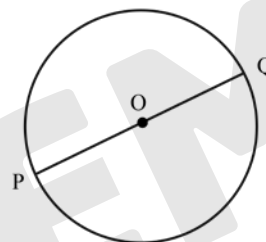
Chord

Chord: If we take two points P and Q on a circle, then the line segment PQ is called a chord of the circle.



Diameter

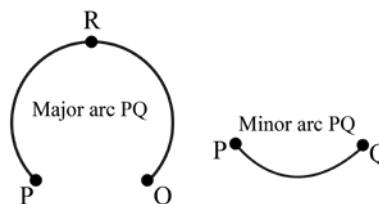
The chord that passes through the centre of the circle, is called the diameter of the circle.



A diameter is the longest chord and all diameters of a circle have the same length, which is equal to two times the radius. In figure, AOB is a diameter of the circle.

Arc

A piece of a circle between two points is called an arc. The longer one is called the major arc PQ and the shorter one is called the minor arc PQ . The minor arc PQ is also denoted by $\widehat{PQ}$ and the major arc QP by $\widehat{QP}$. When P and Q are ends of a diameter, then both arcs are equal and each is called a semi-circle.

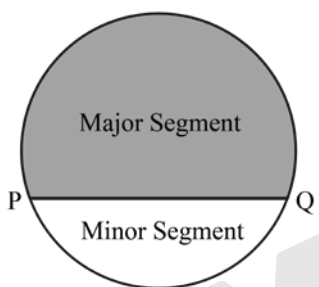


Circumference

The length of the complete circle is called its circumference.

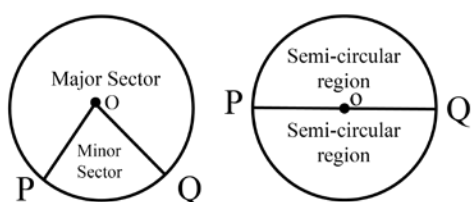
Segment

The region between a chord and either of its arcs is called a segment of the circular region or simply a segment of the circle. There are two types of segments, the major segment and the minor segment (as in figure).



Sector

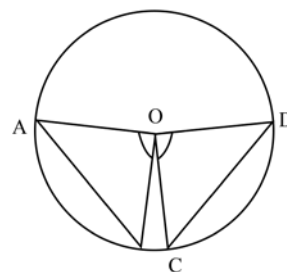
The region between an arc and the two radii joining the centre to the end points of the arc is called a sector. Minor arc corresponds to the minor sector and the major arc corresponds to the major sector. When two arcs are equal, then both segments and both sectors become the same and each is known as a semi-circular region.



Theorems Related to Chords

Theorem

Equal chords of a circle subtend equal angles at the centre.

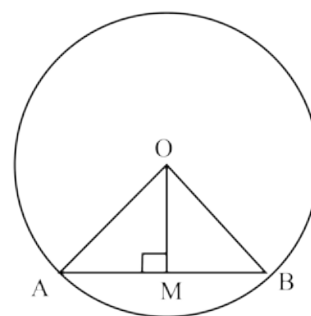


Converse

If the angles subtended by the chords of a circle at the centre are equal, then the chords are equal.

Theorem

The perpendicular from the centre of a circle to a chord bisects the chord.



Converse

The line drawn through the centre of a circle to bisect a chord is perpendicular to the chord.



1. *The radius of a circle is 25 cm and the length of one of its chords is 14 cm. Find the distance of the chord from the centre.



Theorem: There is one and only one circle passing through three given non-collinear points.

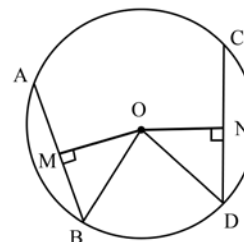


2. *The centre of the circle passing through the midpoints of the sides of an isosceles triangle ABC lies on the circumcircle of triangle ABC . If the larger angle of triangle ABC is α and the smaller one β , then what is the value of $\alpha - \beta$?

[PRMO 2019]



Theorem: Equal chords of a circle (or of congruent circles) are equidistant from the centre (or centres).



Converse: Chords equidistant from the centre of a circle are equal in length.

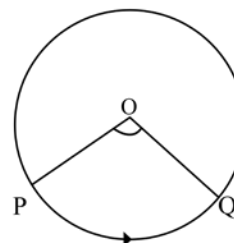


3. AB and CD are equal chords of a circle whose centre is O . When produced, these chords meet at E . Prove that $EB = ED$.



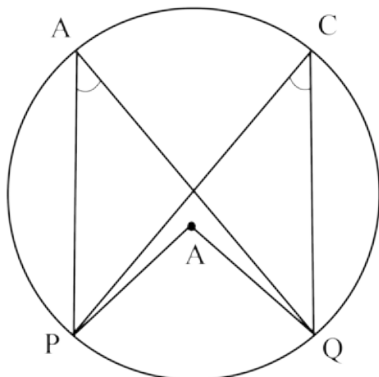
Angle Subtended by an Arc of a Circle

In figure, the angle subtended by the minor arc PQ at O is $\angle POQ$ and the angle subtended by the major arc PQ at O is reflex angle $\angle POQ$.



Theorem: The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

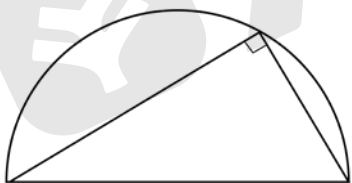
Theorem: Angles in the same segment of a circle are equal.



$$\angle PAQ = \angle PCQ$$

Thales's Theorem

Angle in the semi-circle is a right angle.

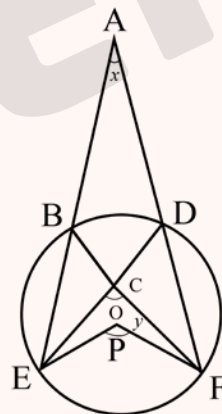


4. In trapezoid $ABCD$ with $\overline{BC} \parallel \overline{AD}$, let $BC = 1000$ and $AD = 2008$. Let $\angle A = 37^\circ$, $\angle D = 53^\circ$, and M and N be the midpoints of $\overline{BC}$ and $\overline{AD}$, respectively. Find the length MN .

[AIME]

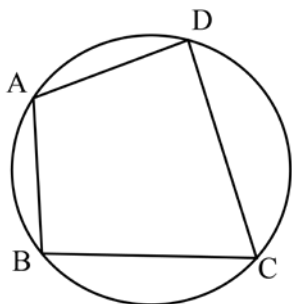


5. In figure, O is the centre of the circle. Prove that $\angle x + \angle y = \angle z$.



Cyclic Quadrilateral

A quadrilateral $ABCD$ is called cyclic if all the four vertices of it lie on a circle.



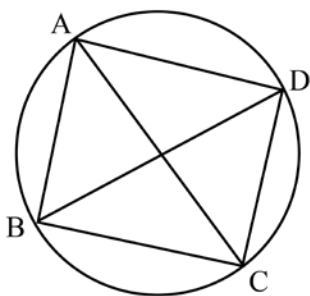
Theorem: The sum of either pair of opposite angles of a cyclic quadrilateral is 180° .

$$\angle A + \angle C = \angle B + \angle D = 180^\circ$$

Theorem: The exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.

Ptolemy's Theorem

In cyclic quadrilateral, the product of the diagonals is equal to the sum of the product of the opposite sides.



$$AC \times BD = AB \times DC + BC \times AD$$

Some important points

1. If the diagonals of a cyclic quadrilateral are diameters of the circle passing through its vertices, then it is a rectangle.
2. If the non-parallel sides of a trapezium are equal, then it is cyclic.
3. Every cyclic parallelogram is a rectangle.
4. The quadrilateral formed by angle bisectors of a cyclic quadrilateral $ABCD$ is also cyclic.
5. If a pair of opposite sides of a cyclic quadrilateral are equal, then its diagonals are equal.
6. If all the sides of a parallelogram touch a circle, then the parallelogram is a rhombus.



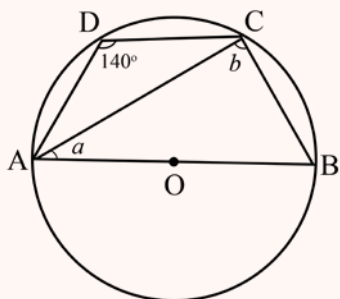
6. A hexagon is inscribed in a circle. Five of the sides have length 81 and the sixth, denoted by $\overline{AB}$, has length 31. Find the sum of the lengths of the three diagonals that can be drawn from A.

[AIME]





7. In the given figure, $AD = DC$ and O is the centre of circle. Find the value of a and b .

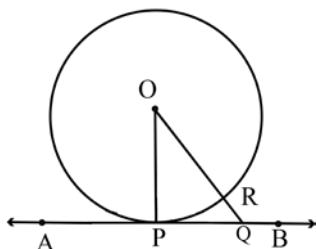


Secant and Tangent

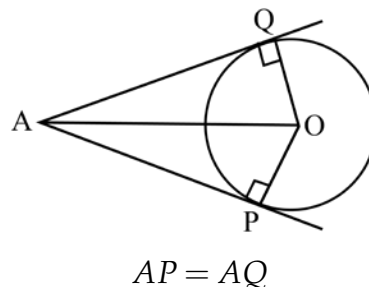
Secant to a circle is a line that intersects the circle in two distinct points.

A tangent to a circle is a line that touches the circle at exactly one point.

Theorem: A tangent to a circle is perpendicular to the radius through the point of contact.



Theorem: Lengths of two tangents drawn from an external point to a circle are equal.



Note:

- ❖ If two tangents are drawn to a circle from an external point, then they subtend equal angles at the centre.

$$\angle QOA = \angle POA \quad [\text{By CPCT}]$$

- ❖ If two tangents are drawn to a circle from an external point, they are equally inclined to the segment, joining the centre to that point

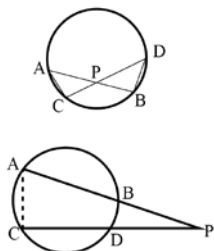
$$\angle OAQ = \angle OAP \quad [\text{By CPCT}]$$

Segment of a Chord

Let AB be a chord of a circle and let P be a point on AB inside the circle. Then, P is said to divide AB internally into two segments PA and PB .



Theorem: If two chords of a circle intersect inside or outside the circle when produced, the rectangle formed by two segments of one chord is equal in area to the rectangle formed by the two segments of another chord.



$$PA \times PB = PC \times PD$$



8. *Two chords AB and CD of a circle intersect each other at O internally.

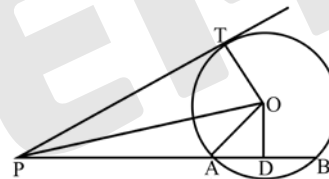
If $AO = 3.5$ cm, $CO = 5$ cm and $DO = 7$ cm, find OB .



Secant Tangent Theorem

If PAB is a secant to a circle intersecting the circle at A and B and PT is a tangent segment, then

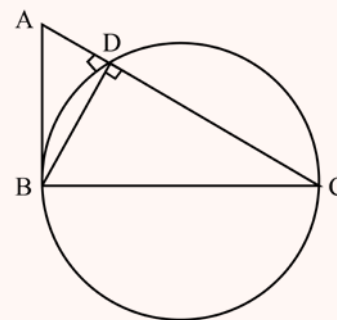
$$PA \times PB = PT^2$$



10. In a right triangle ABC , the perpendicular BD on the hypotenuse AC is drawn. Prove that

(i) $AC \times AD = AB^2$

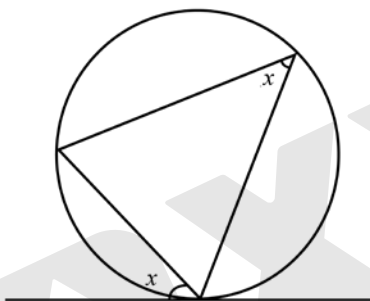
(ii) $AC \times CD = BC^2$



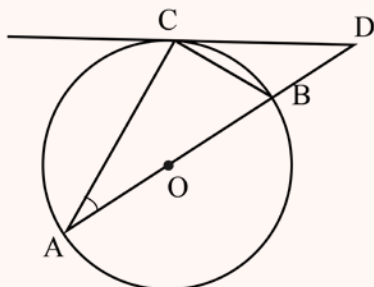
9. In $\triangle ABC$, $AP \perp BC$ and $BQ \perp AC$, which intersect each other at O . Prove that $AO \times OP = BO \times OQ$.



Theorem: If a tangent touches a circle at a point and a chord is drawn from that point of contact, then the angles formed by the chord and the tangent are equal to the angles in the alternate segments.



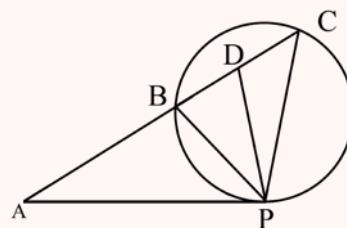
11. AB is diameter and AC is a chord of a circle such that $\angle BAC = 30^\circ$. The tangent at C intersects AB produced at D . Prove that $BC = BD$.



12. In the adjoining figure, AP is a tangent to the circle at P . ABC is a secant such that PD is the bisector of $\angle BPC$.

Prove that

- (i) $\angle ADP = \angle APD$
- (ii) $AP = AD$
- (iii) $\angle BPD = \frac{1}{2}(\angle ABP - \angle APB)$





13. *Let ABC be a triangle such that $AB = AC$. Suppose the tangent to the circumcircle of ABC at B is perpendicular of AC . Find $\angle ABC$ measured in degrees.

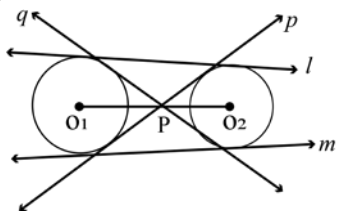
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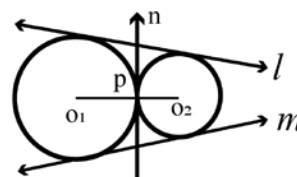
Common Tangents to Two Circles

A line which touches two given circles is called common tangent to the two circles. Let $C(O_1, r_1), C(O_2, r_2)$ be two given circles. Let the distance between centres O_1 and O_2 be d i.e., $O_1O_2 = d$.

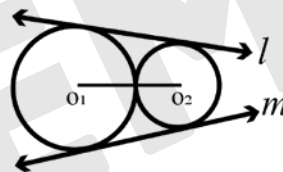
- ❖ In fig.(i) $d > r_1 + r_2$ i.e., two circles do not intersect. In this case, four common tangents are possible. The tangent lines ℓ and m are called direct common tangents and the tangent lines p and q are called indirect (transverse) common tangents.



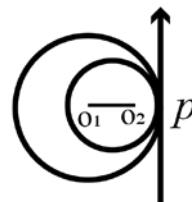
- ❖ In fig.(ii), $d = r_1 + r_2$. In this case, two circles touch externally and there are three common tangents.



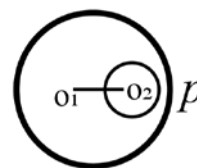
- ❖ In fig.(iii), $d < r_1 + r_2$. In this case two circles intersect in two distinct points and there are only two common tangents.



- ❖ In fig.(iv), $d = r_1 - r_2$ ($r_1 > r_2$), in this case, two circles touch internally and there is only one common tangent.



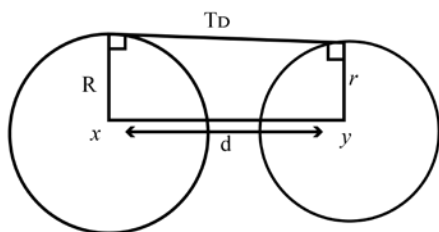
- ❖ In fig.(v), the circle $C(O_2, r_2)$ lies wholly in the circle $C(O_1, r_1)$ and there is no common tangent.



Some important points

Length of the direct common tangent

$$T_D = \sqrt{d^2 - (R - r)^2}$$



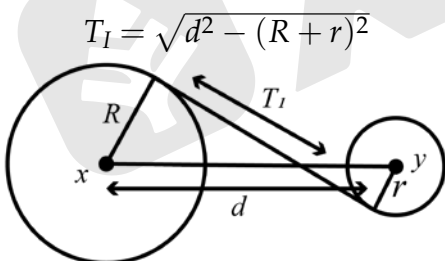
Note:

If two circles touch each other externally then

$$d = R + r$$

$$T_D = 2\sqrt{R \cdot r}$$

Length of transverse common tangent



Note:

If two circles touch externally then

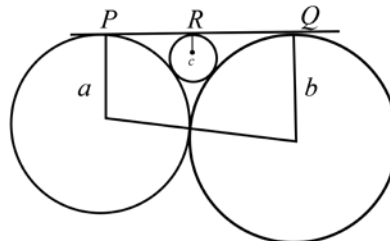
$$d = R + r$$

$$\therefore T_I = \sqrt{(R + r)^2 - (R + r)^2} = 0$$

i.e length of transverse common tangents is zero.

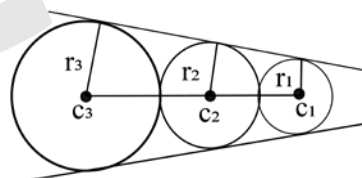
Two circles with radii a and b respectively touch each other externally. Let C be the radius of a circle that touches these two circles as well as a common tangent to the two circles, then

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$$



Three circles with centres C_1, C_2, C_3 with radii r_1, r_2, r_3 where $(r_1 > r_2 > r_3)$ are placed as shown in figure then,

$$r_2 = \sqrt{r_1 r_3}$$



14. *Two circles touch each other externally. Their radii are 9 cm and 4 cm and the distance between their centres is 13 cm. Find the length of their common tangent segment.



Power of a point

The Power of a Point Theorem is a concept that relates a point to a circle. It quantifies how far a point is from a circle, and it applies whether the point is inside, on, or outside the circle.

The Power of a Point P with respect to a circle is defined as:

$$\text{Power of } P = (\text{Distance from } P \text{ to center})^2 - (\text{radius})^2$$

Secant - Secant Case:

If you draw two segments from point P intersecting the circle at points A, B and C, D , then:

$$PA \cdot PB = PC \cdot PD$$

This product is the power of the point.

Secant Tangent Case:

If P is outside the circle and you draw a tangent from P to the circle (touching at point T), and a secant cutting the circle at points A and B :

$$PT^2 = PA \cdot PB$$

This shows that the square of the tangent length equals the power of the point.

Location of Point P	Power of the Point	Example
Outside the circle	Positive	Tangents exist
On the circle	Zero	$OP = r$
Inside the circle	Negative	No real tangents

Radical Axis of Two Circles

The radical axis of two circles is the locus of all points that have equal power with respect to both circles.

That is, for any point P on the radical axis:

$$\begin{aligned} \text{Power of } P \text{ with respect to circle 1} \\ = \text{Power of } P \text{ with respect to circle 2} \end{aligned}$$

Geometric Interpretation:

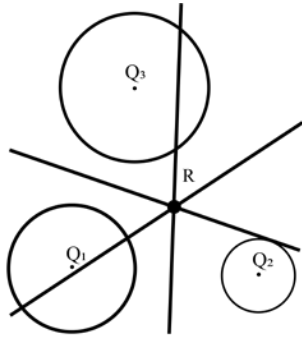
- ❖ The radical axis is always a straight line.
- ❖ If the circles intersect, the radical axis is the line through the points of intersection.
- ❖ If the circles do not intersect, the radical axis lies between or outside them, depending on the sizes.

Radical Centre of Three Circles

The radical centre is the common point of intersection of the radical axes of any three circles (assuming the centres are not collinear).

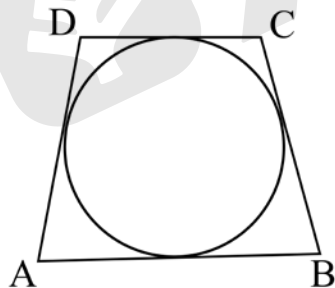
- ❖ Take three circles: C_1, C_2, C_3
- ❖ Form three radical axes:
 $\text{Rad}(C_1, C_2), \text{Rad}(C_2, C_3), \text{Rad}(C_1, C_3)$
- ❖ These three lines intersect at a single point - this point is the radical centre.

- ❖ The radical centre is equidistant in power from all three circles.



Tangential Quadrilateral

A convex quadrilateral whose sides are all tangent to a single circle within the quadrilateral is called a tangential quadrilateral (or circumscribed quadrilateral or inscriptible quadrilateral). This circle is called the incircle of the quadrilateral or its inscribed circle, its center is called the incenter and its radius is called the inradius.



Pitot Theorem

Let $ABCD$ be a tangential quadrilateral then the sum of the opposite sides are equal that is,
 $AB + CD = AD + BC$

Converse of Pitot Theorem

Any convex quadrilateral that satisfies
 $AB + CD = AD + BC$ is tangential.



15. If all the sides of a parallelogram touches a circle, show that the parallelogram is a rhombus.



Area Related to Circles

For any circle,

$$\frac{\text{Circumference}}{\text{Diameter}} = \pi$$

If r is the radius of a circle, then

- ❖ Circumference $= 2\pi r$ or πd , where
 $d = 2r$ is the diameter of the circle.



❖ Area = πr^2 .

❖ Area of semi-circle = $\frac{\pi r^2}{2}$.

❖ Perimeter of the semi-circle = $\pi r + 2r$.

❖ Area enclosed by two concentric circles
 $= \pi R^2 - \pi r^2 = \pi (R^2 - r^2)$
 $= \pi (R + r)(R - r)$

where R and r are radii of two concentric circles.



16. Find the area of a circle whose circumference is 22 cm.



Area of a Sector

If A be the area of the sector, then $A = \frac{\pi r^2 \theta}{360^\circ}$
 and length of an arc $\ell = \frac{\pi r \theta}{180^\circ}$, where θ is the central angle of the sector.



Some Important Points

❖ Angle described by minute hand in 60 minutes = 360° .

$\therefore$ Angle described by minute hand in one minute = $\frac{360^\circ}{60} = 6^\circ$.

Thus, minute hand rotates through an angle of 6° in one minute.

❖ Angle described by hour-hand in 12 hours. = 360° .

$\therefore$ Angle described by hour hand in one hour = $\left(\frac{360^\circ}{12}\right)^\circ = 30^\circ$

$\therefore$ Angle described by hour hand in one minute = $\frac{30^\circ}{60} = \left(\frac{1}{2}\right)^\circ$.

Thus, hour hand rotates through an angle of $\left(\frac{1}{2}\right)^\circ$ in one minute.



17. A sector is cut from a circle of radius 21 cm. The angle of the sector is 150° . Find the length of its arc and its area.



Important Formulas

❖ Heron's formula: Area of a triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$
 where s = semi-perimeter and a, b, c are the sides of the triangle.

❖ Area of a right-angled triangle

$$= \frac{1}{2} \times \text{base} \times \text{altitude}$$

❖ Area of an equilateral triangle $= \frac{\sqrt{3}}{4}a^2$.

❖ Area of a rectangle = length $\times$ breadth

❖ Perimeter of a rectangle

$$= 2(\text{length} + \text{breadth})$$

❖ Area of a square of side ' a ' $= a^2$.

❖ Length of the diagonal of a square of side ' a ' $= \sqrt{2}a$.

❖ Perimeter of a square of side ' a ' $= 4a$.

❖ Area of a parallelogram = base $\times$ height

❖ Area of a rhombus $= \frac{1}{2}d_1d_2$.
 where d_1 and d_2 are the lengths of its diagonals.

❖ Area of a trapezium $= \frac{1}{2}(a+b)h$
 Where a and b are lengths of the two parallel sides and h is the distance between them.

❖ With sides of a cyclic quadrilateral being a, b, c, d and semi-perimeter s , the area K of a cyclic quadrilateral is

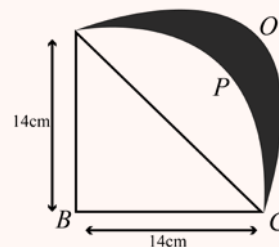
$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$



18. The diameter of a cycle wheel is 28 cm. How many revolutions will it make in moving 13.2 km?

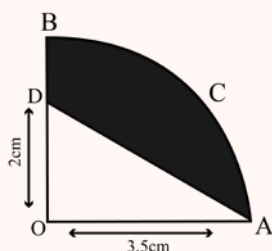


19. *In the given figure $ABCP$ is a quadrant of a circle of radius 14 cm. With AC as diameter, a semi-circle is drawn. Find the area of the shaded portion.





20. In figure, $AOBCA$ represents a quadrant of a circle of radius 3.5 cm with centre O . Calculate the area of the shaded portion (take $\pi = \frac{22}{7}$)



Polygon

A closed plane figure bounded by line segments is called a polygon. A polygon is named according to the number of sides it has:

No. of sides	3	4	5	6	7	8	10
Figure	Triangle	Quadrilateral	Pentagon	Hexagon	Heptagon	Octagon	Decagon

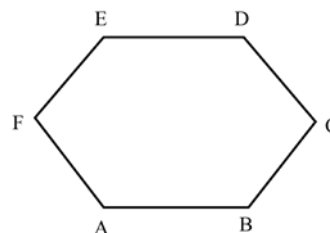
In general, a polygon having n sides is called n -sided polygon.

Diagonal of Polygon

Line segment joining any two non-consecutive vertices of a polygon is called its diagonal.

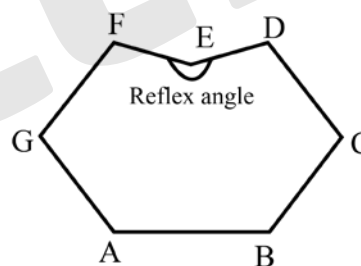
Convex Polygon

If all the interior angles of a polygon are less than 180° , it is called a convex polygon.



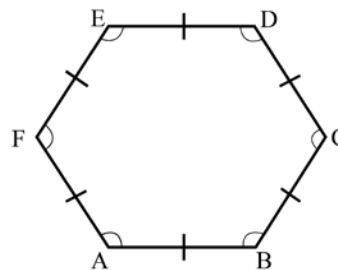
Concave Polygon

If one or more of the interior angles of a polygon is greater than 180° (i.e., reflex), it is called a concave polygon.



Regular Polygon

A polygon is called a regular polygon if all its sides have equal length and all the angles have equal measure.



Note

- ❖ The sum of the interior angles of a convex polygon of n sides is $(2n - 4)$ right angles or $(2n - 4) \times 90^\circ$.
- ❖ The sum of the exterior angles of a polygon is 4 right angles or 360° .
- ❖ Each interior angle of a n -sided regular polygon is $\frac{(2n - 4) \times 90^\circ}{n}$.
- ❖ Each exterior angle of a regular polygon of n sides $= \left(\frac{360^\circ}{n}\right)$.
- ❖ If a polygon has n sides, then the number of diagonals of the polygon $= \frac{n(n - 3)}{2}$.

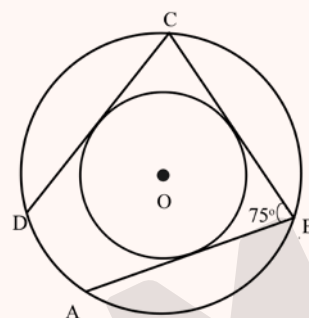


21. *Let X be the set of all even positive integers n such that the measure of the angle of some regular polygon is n degrees. Find the number of elements in X .

[IOQM 2023]



22. *Given a pair of concentric circles, chords $AB, BC, CD, \dots$ of the outer circle are drawn such that they all touch the inner circle. If $\angle ABC = 75^\circ$, how many chords can be drawn before returning to the starting point?

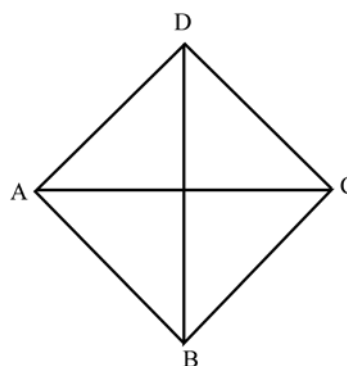


[IOQM 2021]



Quadrilaterals

A quadrilateral is a four-sided closed figure.



Let A , B , C and D be four points in a plane such that:

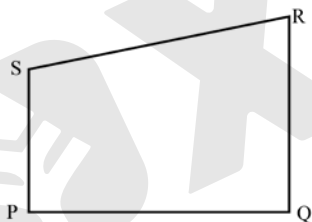
- (i) No three of them are collinear.
- (ii) The line segments AB , BC , CD , and DA do not intersect except at their end points,

then figure obtained by joining A , B , C and D is called a quadrilateral.

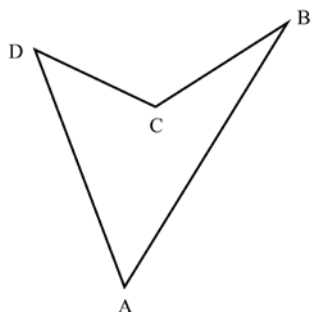
Convex and Concave Quadrilaterals

- ❖ A quadrilateral in which the measure of each interior angle is less than 180° is called a convex quadrilateral.

In figure, $PQRS$ is a convex quadrilateral.



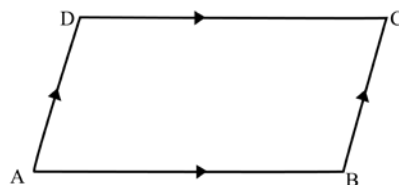
- ❖ A quadrilateral in which the measure of one of the interior angles is more than 180° is called a concave quadrilateral. In figure, $ABCD$ is a concave quadrilateral.



Special Quadrilateral

Parallelogram

A parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. In figure, $AB \parallel DC$, $AD \parallel BC$. Therefore, $ABCD$ is a parallelogram.



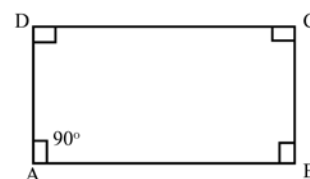
Properties:

- ❖ A diagonal of a parallelogram divides it into two congruent triangles.
- ❖ In a parallelogram, opposite sides are equal.
- ❖ The opposite angles of a parallelogram are equal.
- ❖ The diagonals of a parallelogram bisect each other.

Rectangle

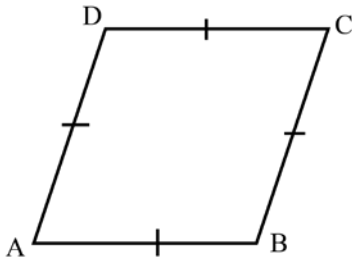
A rectangle is a parallelogram, in which each of its angle is a right angle. If $ABCD$ is a rectangle, then

$\angle A = \angle B = \angle C = \angle D = 90^\circ$, $AB = CD$, $BC = AD$, and diagonals $AC = BD$.



Rhombus

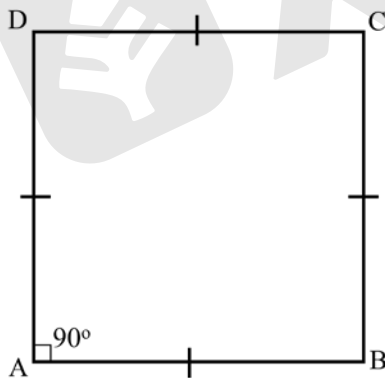
A rhombus is a parallelogram in which all its sides are equal in length. If $ABCD$ is a rhombus, then $AB = BC = CD = DA$.



The diagonals of a rhombus are perpendicular to each other.

Square

A square is a parallelogram having all sides equal and each angle equal to right angle. If $ABCD$ is a square, then $AB = BC = CD = DA$, diagonal $AC = BD$, and $\angle A = \angle B = \angle C = \angle D = 90^\circ$.

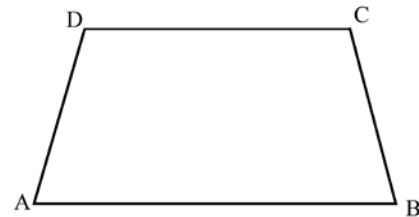


The diagonals of a square are perpendicular to each other.



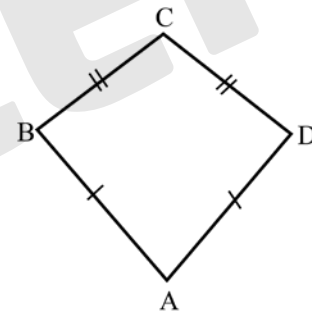
Trapezium

A trapezium is a quadrilateral with only one pair of opposite sides parallel. In figure, $ABCD$ is a trapezium with $AB \parallel DC$.



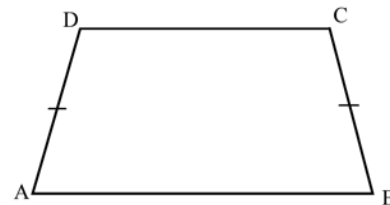
Kite

A kite is a quadrilateral in which two pairs of adjacent sides are equal. If $ABCD$ is a kite then $AB = AD$ and $BC = CD$.



Isosceles trapezium

A trapezium is said to be an isosceles trapezium, if its non-parallel sides are equal. Thus, a quadrilateral $ABCD$ is an isosceles trapezium, if $AB \parallel DC$ and $AD = BC$.



In an isosceles trapezium $\angle A = \angle B$ and $\angle C = \angle D$.

Properties

Theorem 1: The sum of the four angles of a quadrilateral is 360° .

Theorem 2: A diagonal of a parallelogram divides it into two congruent triangles.

Theorem 3: In a parallelogram, opposite sides are equal.

Theorem 4: The opposite angles of a parallelogram are equal.

Theorem 5: The diagonals of a parallelogram bisect each other.

Theorem 6: Each of the four angles of a rectangle is a right angle.

Theorem 7: Each of the four sides of a rhombus is of the same length.

Theorem 8: Each of the angles of a square is a right angle and each of the four sides is of the same length.

Theorem 9: The diagonals of a rectangle are of equal length.

Theorem 10: The diagonals of a rhombus are perpendicular to each other.

Theorem 11: The diagonals of a square are equal and perpendicular to each other.

A quadrilateral becomes a parallelogram when:

- ❖ Opposite angles are equal.
- ❖ Both pairs of opposite sides are equal
- ❖ A pair of opposite side is equal as well as parallel
- ❖ Diagonals bisect each other.

Note:

- ❖ Square, rectangle, and rhombus are all parallelograms.
- ❖ Kite and trapezium are not parallelograms.
- ❖ A square is a rectangle.
- ❖ A square is a rhombus.
- ❖ A parallelogram is a trapezium.

Varignon's Theorem

The quadrilateral formed by joining the mid point of consecutive sides of any quadrilateral is a parallelogram.



Varignon's Corollary: Rectangle

The quadrilateral formed by joining the mid points of consecutive sides of quadrilateral whose diagonals are perpendicular is a rectangle.

Varignon's Corollary: Rhombus

The quadrilateral formed by joining the mid points of consecutive sides of quadrilateral whose diagonals are congruent is a rhombus.

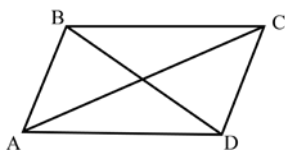
Varignon's Corollary: Square

The quadrilateral formed by joining the mid points of consecutive sides of quadrilateral whose diagonals are congruent and perpendicular is a square.

Some Important Points

1. The line segment joining the mid points of the opposite sides of any quadrilateral bisect each other.
2. The sum of the squares of the lengths of the sides of a parallelogram equals the sum of the squares of the lengths of the diagonals.

$$AB^2 + BC^2 + CD^2 + AD^2 = AC^2 + BD^2$$

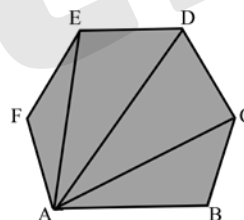


3. The angle bisectors of a parallelogram form a rectangle.
4. The angle bisectors of a rectangle form a square.

Areas of Polygonal Regions

Polygon Region

A polygon region can be expressed as the union of a finite number of triangular regions in a plane such that if two of these intersect, their intersection is either a point or a line segment. It is the shaded portion including its sides as shown in the figure.



Area Axioms

Every polygonal region R has an area, measured in square units and denoted by $ar(R)$.

- ❖ **Congruent area axiom:** If R_1 and R_2 be two regions such that $R_1 \cong R_2$, then $ar(R_1) = ar(R_2)$.
- ❖ **Area addition axiom:** If R_1 and R_2 are two polygonal regions, whose intersection is a finite number of points and line segments such that $R = R_1 \cup R_2$, then $ar(R) = ar(R_1) + ar(R_2)$.

❖ **Rectangular area axiom:**

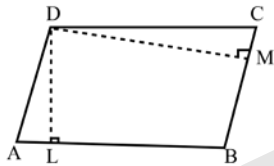
If $AB=a$ metre and $AD=b$ metre then,
 $ar(\text{rectangular region } ABCD)$
 $=ab$ sq.m.

Area of a Parallelogram

Base and altitude of a parallelogram:

Base: Any side of a parallelogram can be called its base.

Altitude: The length of the line segment that is perpendicular to the base from the opposite side is called the altitude or height of the parallelogram corresponding to the given base.



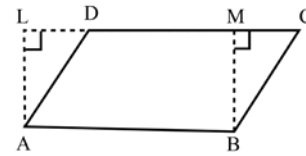
- DL is the altitude of parallelogram $ABCD$, corresponding to the base AB .
- DM is the altitude of parallelogram $ABCD$, corresponding to the base BC .

Theorem: A diagonal of a parallelogram divides it into two triangles of equal area.

Theorem: Parallelograms on the same base and between the same parallels are equal in area.



Theorem: The area of parallelogram is the product of its base and the corresponding altitude.



$$\text{Area} = AB \times AL$$

Area of a Triangle

Theorem: Two triangles on the same base (or equal bases) and between the same parallels are equal in area.

Area of a Trapezoid

A trapezoid (or trapezium) has one pair of parallel sides, called bases.

Formula:

$$\text{Area} = \frac{1}{2} \times (b_1 + b_2) \times h$$

- b_1 = length of the first base
- b_2 = length of the second base
- h = height (perpendicular distance between the bases)



Area of a Regular Hexagon

A regular hexagon can be divided into 6 equilateral triangles.

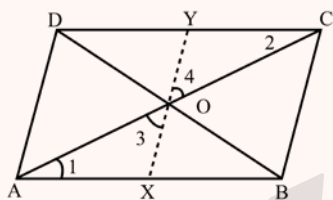
Formula:

$$\text{Area} = \frac{3\sqrt{3}}{2} \times s^2$$

- s = side length of the hexagon



23. The diagonals of a parallelogram $ABCD$ intersect at O . A line through O intersects AB at X and DC at Y . Prove that $OX = OY$



24. In a parallelogram $ABCD$, $AB = 8$ cm. The altitudes corresponding to sides AB and AD are, respectively, 4 cm and 5 cm. Find AD .



25. $ABCD$ is a trapezium with $AB \parallel DC$. A line parallel to AC intersects AB at X and BC at Y . Prove that $\text{ar}(\triangle ADX) = \text{ar}(\triangle ACY)$.



26. *Let $ABCD$ be a convex cyclic quadrilateral. Suppose P is a point in the plane of the quadrilateral such that the sum of its distances from the vertices of $ABCD$ is the least. If $\{PA, PB, PC, PD\} = \{3, 4, 6, 8\}$. What is the maximum possible area of $ABCD$?

[PRMO 2019]



27. *Let D be an interior point of the side BC of a triangle ABC . Let I_1 and I_2 be the incentres of triangles ABD and ACD , respectively. Let AI_1 and AI_2 meet BC in E and F , respectively. If $\angle BI_1E = 60^\circ$. What is the measure of $\angle CI_2F$ in degrees?

[PRMO 2018]

29. Triangle ABC has $AB = 21, AC = 22$ and $BC = 20$. Points D and E are located on $\overline{AB}$ and $\overline{AC}$, respectively, such that $\overline{DE}$ is parallel to $\overline{BC}$ and contains the center of the inscribed circle of triangle ABC . Then $DE = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

[AIME]



28. *In convex hexagon $ABCDEF$, all six sides are congruent, $\angle A$ and $\angle D$ are right angles, and $\angle B, \angle C, \angle E$ and $\angle F$ are congruent. The area of the hexagonal region is $2116(\sqrt{2} + 1)$. Find AB .

[AIME]

30. *Let $ABCD$ be a rectangle in which $AB + BC + CD = 20$ and $AE = 9$ where E is the mid-point of the side BC . Find the area of the rectangle.

[IOQM 2021]



31. *Let $ABCD$ be a trapezium in which $AB \parallel CD$ and $AB = 3CD$. Let E be then mid-point of the diagonal BD . If $[ABCD] = n \times [CDE]$, what is the value of n ? (Here $[\tau]$ denotes the area of the geometrical figure τ).

[IOQM 2021]

ANSWER KEY

(1) 24	(9) -	(17) length of arc = 55 area = 577.5	(24) 6.4
(2) 90°	(10) -	(18) 15000	(25) -
(3) -	(11) -	(19) 98	(26) 55
(4) 504	(12) -	(20) 6.125	(27) 30
(5) -	(13) 30	(21) 16	(28) 46
(6) 384	(14) 12	(22) 24	(29) 923
(7) $a = 50$ $b = 90$	(15) -	(23) -	(30) 19
(8) 10	(16) 38.5		(31) 8

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